



ALL-AFRICA STUDENTS UNION
UNION PANAFRICAINE DES ETUDIANTS
الاتحاد جميع الطلاب الأفريقي
Defending students rights since 1972

CONCEPT NOTE FOR GIRLS IN STEM



1.0 BACKGROUND AND JUSTIFICATION (EDUCATION OF GIRLS and WOMEN IN SCIENCE, TECHNOLOGY, ENGINEERING AND MATHEMATICS) (STEM)

In recent times, girls are in school than ever before, but they do not always have the same identical opportunities as boys to finish and benefit from their choice of college. Social norms and perceptions that affect the standard of the education they receive and the subjects they research keep so many girls and women back. They are especially under-represented in education in science, technology, engineering and mathematics (STEM), and thus in careers in STEM.

According to the UNESCO's groundbreaking study "Cracking the code: Girls' and women's education in STEM", just 35 percent of STEM students in higher education worldwide are women, and disparities are observed within STEM disciplines.

The ability and talent of girls in STEM has not, in many respects been acknowledged and embraced culturally and politically. This definitely poses a big problem and also a big challenge not only to ensure the availability of much needed STEM quality staff themselves, but also to provide many great opportunities for career growth in STEM that girls and women also deserve.

Girls and women need an education that goes deeper than literacy and numeracy which equips them with instruments to engage fairly and actively in solving the dynamic problem facing our world. With the acceleration of technological transition and economies changing quickly to face the consequences of the new Global pandemic COVID-19, failure to invest in girls' STEM education, and empowerment of women and girls to participate in digital transition, will forfeit half of the population's economic capacity, yet exempt world-class girls and women from the digital learning applications and an entire set of solutions and a whole range of new occupations that require STEM-related skills.

PROJECT OBJECTIVES

- This project seeks to build the interest of young people (Girls) in the continent in Science, Technology, Engineering and Mathematics. (STEM). Course and work occupation in the areas of Science, Technology, engineering and Mathematics are mostly assumed to be man's field and male dominated.
- This project seeks to break this barrier and assumption and encourage girls in schools to choose courses in the sciences, Technology, Engineering and mathematics and to maintain interest in them and in future choose job occupation on these fields.
- This project also seeks to show young people (Girls) they can be just as successful as boys in Science, Technology, Engineering and Mathematics (STEM) fields. Girls in the subjects are just as qualified as boys are. The difference is that they have to realize the STEM is for them. All goes back to standards.
- This project seeks to Strengthen countries' ability to provide gender-responsive STEM education, including through teacher education, educational content and pedagogy.
- Strengthen awareness of the importance for girls and women of STEM education.

PROJECT ACTIVITIES

- Boot camps to teach the girls coding and robotics
- Seminars and Workshops that will bring girls into direct contact with mentors who will mentor them over the six months period.
- Reviewing of existing STEM curriculum to incite more girls into STEM.
- Establishing clubs in schools to keep the passion of girls for STEM alive.

PROJECT OUTCOMES

It is expected that, the project will offer effective and promising strategies for promoting diverse girls' opportunities in STEM, leveling the girls' playing field and resolving workforce shortages. These strategies will be led and determined by AASU to review a wide array of published materials, interviewed leaders of nationally recognized STEM initiatives.

The long-term outcome will be the reduction on national unemployment figures of young people (girls) and the improvement in the growth of the national economy. The project will encourage young people (Girls) in the continent to have interest and pursue courses in the areas of Science, Technology, engineering and Mathematics (STEM) to secure future jobs on these fields.

A revised version of the existing curriculum that meets the current needs of girls. Thus, practicable and fun curriculum that will increase the number of girls interested in STEM.

Also, women and girls will excel in STEM fields and become role models for upcoming generations

PARTNERS

- Ghana Education Service
- Ministry of Gender, Children and Social Protection
- Snake Nation
- Butterfly Effect
- Plan International
- Soronko Academy
- Google skills for Africa

PROJECT DURATION

Boot camp for a minimum of one week

Schedule for the project will be tailored to varying academic calendar for targeted countries

Timeline: July 2021 to December 2021

PROJECT BENEFICIARIES

- Women and Girls
- Disabled Groups
- Primary school students
- Secondary school students
- University students